

Welcome to STA304 Summer 2026!

Surveys, Sampling, and Observational Data

Emily Somerset, May 5th, 2026



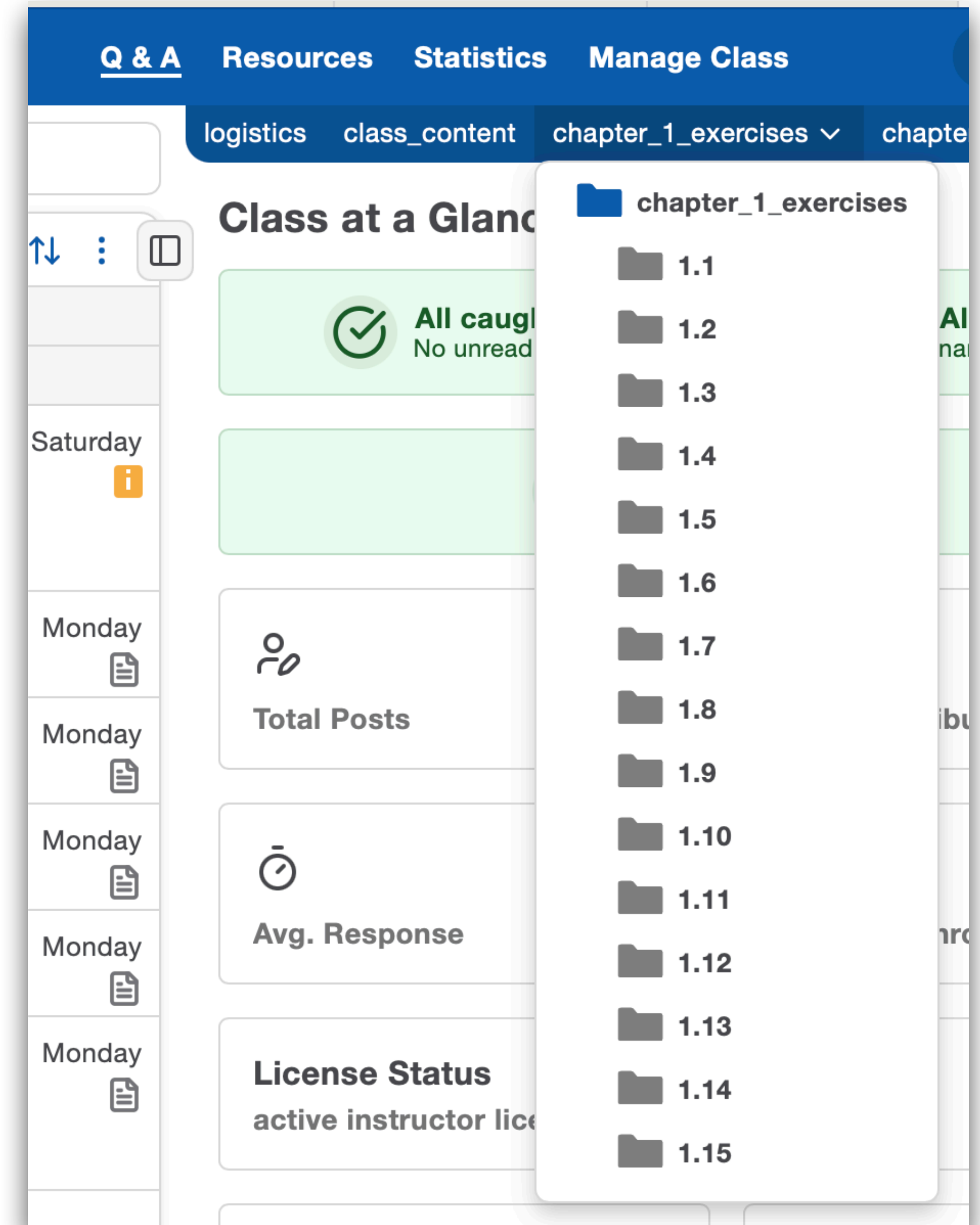
Quercus

- Announcements
- Office hour schedule
- Regrade Request form
- R resources
- Suggested textbook problems.
- Test information (to be posted)
- Lecture notes, lecture notes solutions
- Learning activities
- PDF of textbook



Suggested textbook problems

- Suggested textbook problems (updated after each lecture)
 - We are using a freely available textbook.
 - No solutions available.
- However I will create folders on piazza for ease of answer sharing.
- PDF of textbook posted on Quercus



Supplementary textbooks

All course materials are provided on Quercus. The supplementary textbooks below are:

1. Sharon L. Loh *Sampling: Design and Analysis*. Chapman & Hall/CRC, 2021.
2. Richard L. Scheaffer, William Mendenhall, III, R. Lyman Ott, Kenneth G. Gerow *Elementary Survey Sampling, 7th Edition*. Boston: Pws-Kent, 2011.
3. Nathan Taback *Design and Analysis of Experiments and Observational Studies using R*. Chapman & Hall/CRC, 2022.

1 and 3 are free.

Grading Scheme + Test Schedule

The assessments for STA304 are listed in table below. Your final grade will be calculated as the maximum of Weight Scheme A and Weight Scheme B.

Assessment	Weight scheme A	Weight scheme B	Due Date / Details
Learning activities	Best 8/10 10%	Best 8/10 10%	Generally due Tues/Thurs at 11:59 PM ET (see calendar).
Term Test 1	25%	15%	Thursday, May 21, 2:30 PM – 4:30 PM ET (during class time) • Location TBA on Quercus.
Term Test 2	15%	25%	Thursday, June 4, 2:30 PM – 4:30 PM ET (during class time) • Location TBA on Quercus.
Final Exam	50%	50%	To be scheduled by the Faculty of Arts & Science. • Do not book travel until the official schedule is released. • Conflicts/deferrals: artsci.utoronto.ca.
Total	100%	100%	

Learning Activities

- Learning activities are posted after each class (Tuesdays and Thursdays)
- These are low-stakes formative assessments (to help you learn)
- Submit your work on Quercus.
- Best 8 out of 10 will count towards your final grade.
- Worth 10% in total.

Learning Activity 1

Due May 12 at 11:59p.m.

Points 2

Available after May 5 at 5p.m.

This assignment is locked until May 5 at 5p.m..

Learning Activities				
Criteria	Ratings			Pts
Grade	2 pts Genuine attempt (full marks) • Completes all or most of the activity • Responses are connected to the course material (even if incorrect) • Demonstrates effort to apply relevant concepts	1 pts Minimal engagement • Incomplete (e.g., only part of the activity is attempted) • Responses are extremely vague • Limited evidence of engagement with the material	0 pts Missing or irrelevant • No submission, or response is clearly unrelated to the activity.	2 pts
				Total Points: 2

Learning Activities

- You may use generative AI tools for learning support, but be aware that over-reliance may limit your ability to develop the statistical reasoning skills this course is designed to build.
- It is strongly recommended that you attempt each question earnestly before consulting external tools.
- I do not expect your responses to be perfect (see rubric). Consider these activities as low-stakes opportunities to engage more deeply with course concepts.

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Learning Activities

- Teaching assistant’s will be providing feedback for these learning activities.
- If you would like more feedback (not a re-marking request), please email me.

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				Total Points: 2

Course Calendar

- On syllabus.
- You can find all learning activity deadlines.
- The syllabus will be posted soon. Just waiting for it to be approved.

Date	Lec.	Topics	Deliverables
May 5	L1	• Elements of the sampling problem • Selection bias • Measurement error • Questionnaire design • Ethics in research	LA1 posted
May 7	L2	• Framework for probability sampling • Types of probability samples • Simple random sample (SRS)	LA2 posted
May 12	L3	• Inference under SRS for population mean, total, and proportion. • When SRS should be used	LA3 posted LA1 due LA2 due
May 14	L4	• Inference under stratified sampling (STRS) for population mean, total, and proportion. • Allocating observations to strata	LA4 posted LA3 due
May 19	L5	• Defining Strata • Ratio estimation • Regression estimation	LA5 posted LA4 due

Term tests + Final

- Term test 1, Thursday May 21, 2:30-4:30pm
- Term test 2, Thursday June 4, 2:30-4:30pm
- Final Exam (during exam period)

Preparation

Study Materials

You are strongly encouraged to thoroughly review the following materials in preparation for the midterm.

- Lecture slides
- Practice problem (TO BE POSTED)
consists of relevant questions from previous STA304 tests, recommended textbook questions, professor's questions, and additional multiple choice questions.

Study Halls

*You are strongly encouraged to attend the study halls. Teaching team members including the instructors will be present to answer your questions including any questions regarding the **Practice problem set**.*

Refer to the [Office Hours](#) page for the study halls schedule.

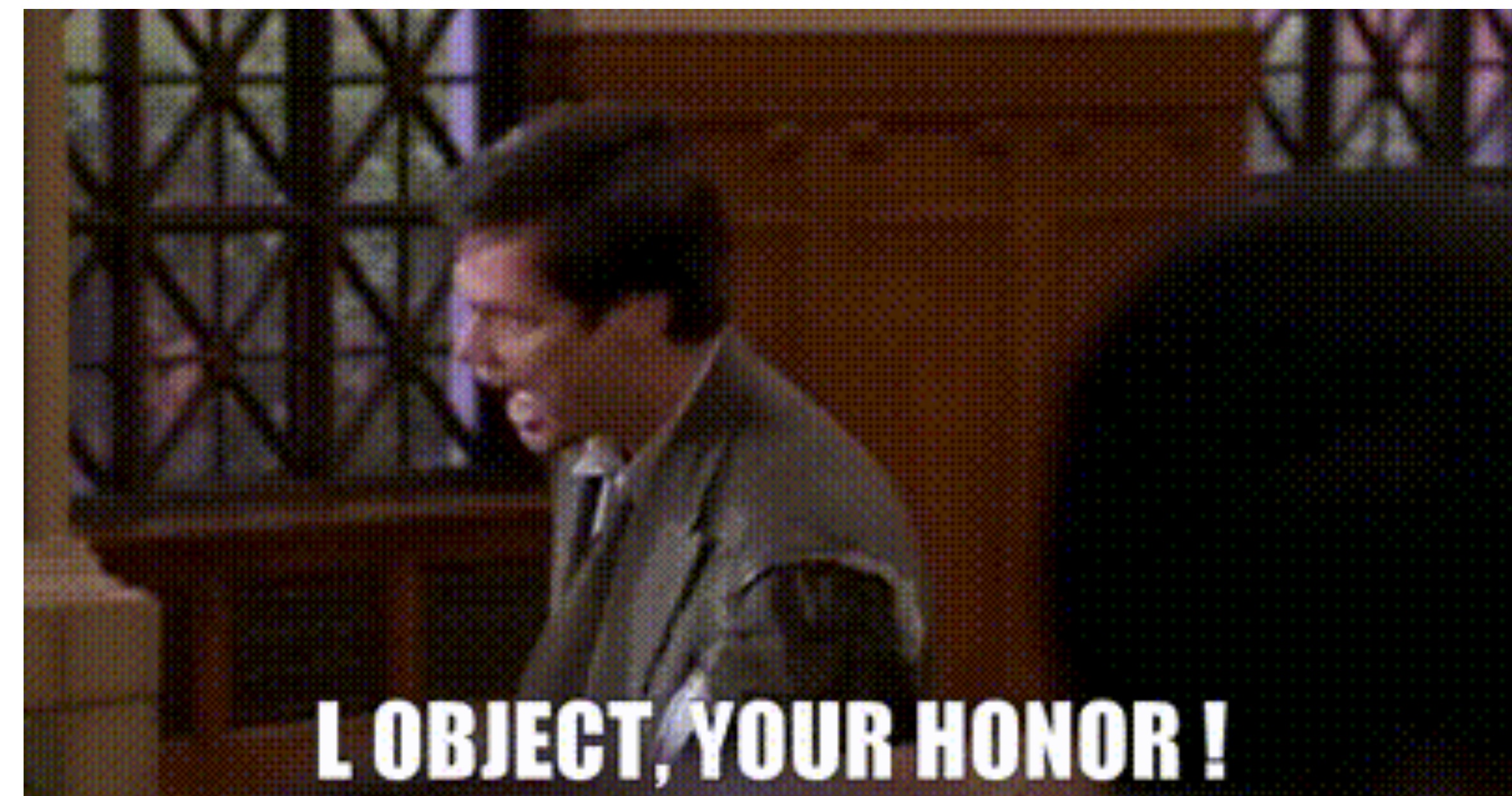
Office Hours

Refer to the [Office Hours](#) page for the office hour schedule.

Note that Friday May 15 and Monday May 18 are holidays and the university is closed. Only the professors office hours on Friday 9am - 10am will be available.

Re-marking policy

- Form on Quercus
- Must not be sent within the first 24 hours of the release of the assessment grade.
- Must be received within one week of the date that the marks for the assessment became available
- Must give specific, clear, and consist reason for each request, referring to a possible error or omission by the marker. Re-mark requests without a specific reason will not be considered.
- All re-marking will be done by the instructor.



General Advice

- I believe that consistent engagement with the material throughout the term is one of the most effective ways to learn.
 - Attend class or watch lecture videos promptly, rather than letting content accumulate.
 - This course will move quickly, so it is important to stay on top of the material as much as possible.

General Advice

- Be thoughtful about how you use generative AI; over-reliance may limit your opportunity to develop your own understanding.
- Try to attempt learning activities or practice problems independently first. The marking scheme is designed to support this approach without penalizing honest effort.
- Consider creating a contingency plan at the start of the semester.
 - “If I get stuck, I will review my notes or lecture examples and identify where my understanding breaks down before seeking outside help.”
 - Or “If I feel tempted to use AI right away, I will spend at least 10–15 minutes attempting the problem on my own first, writing down everything I know.”
 - Or “If I use AI, I will only do so after attempting the problem, and I will treat the response as something to critique and verify, not copy.”

General Advice

- Learning tips
 - UofT offers learning strategist resources that you can access for additional support.
 - Practice explaining the material to a friend (you could also use generative AI for this purpose).
 - Complete the suggested practice problems.
 - Focus on the lecture slides as your primary resource, rather than the textbook

